

SPECIES AND SEX DIFFERENCES IN RESPONSE TO A NOVEL STIMULUS IN *ACOMYS CAHIRINUS* AND *MUS MUSCULUS*

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Summary.—An experiment is described in which the relative distractibility of two species of murid rodent, *Acomys cahirinus* and *Mus musculus*, was compared. On the basis of observations derived from our previous studies of exploratory behaviour in males of these two species, we predicted that *Mus* would show higher levels of distractibility when presented an unexpected, novel stimulus than would *Acomys*. Distractibility was measured using a novel stimulus placed on the wall of a runway along which the animals had been trained to run to a food reward. The hypothesis was not supported, at least for males; indeed, male *Acomys* were more easily distracted from the runway task by introduction of the novel stimulus than male *Mus*. Among females, however, the species difference was reversed for female *Mus* were highly responsive to the novel stimulus. The most significant difference shown in this experiment appears, however, to be that between male and female *Mus*, since male *Mus* showed relatively little response to the stimulus change. The results are discussed in relation to possible hormone differences and differences between the species, in responses to novelty.

In a recent study of patterns of exploratory behaviour in rodents (Birke, D'Udine, & Albonetti, 1985), we have described differences in exploratory patterns between Spiny Mice (*Acomys cahirinus*) and the more familiar laboratory mouse *Mus musculus*. *Mus* tended to be more cautious about emerging into an open arena and to make short forays into it, whereas *Acomys* tended to move rapidly into the arena and to make rapid dashes around the periphery. Whilst there, however, *Acomys* appeared to be little distracted by other stimuli in the arena, spending long periods in sustained locomotor activity; *Mus*, by contrast, moved swiftly between available stimuli rather than around the arena edge, appearing to make frequent shifts of attention between stimuli as they did so.

These observations suggested that there might be a difference between these two species in the extent to which they can be distracted from ongoing behaviour by extraneous cues. The manner in which *Acomys* explored the arena suggested that it might be less readily distracted by other cues, such as a novel stimulus, in the arena.

The present experiment was designed to test this hypothesis, using a runway task. Runway tasks have previously been used to test distractibility (and so to provide a measure of attention in animals) in, for example, young chicks

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(Archer, 1974) and in mice (Archer, 1977), and rats (Birke, *et al.*, 1979). Following training in the runway, experimental animals are presented a novel stimulus at some point in the runway or goal area and the effect on behaviour recorded. "Distractibility" is then measured in terms of the ease with which the animal switches its attention from the task in hand (e.g., running the runway) to the novel stimulus. A similar design was used in the present experiment, in which the behaviour of the two species in the runway was compared.

In this experiment, 10 males and 10 females of each species were about 50 to 70 days old at the time at which training began. The *Acomys* were housed in single-sex groups, 4 to 6 to a cage, of dimensions $22 \times 12 \times 18$ cm, on pine shavings. They were fed rat chow and sunflower seeds and had water available *ad libitum*. The *Mus* were B6/CBA stock and were housed in single-sex groups of 6 to 8 with food and water available *ad libitum*. All animals were bred in the Animal House of the Open University and appeared healthy. They were maintained on a reversed lights schedule (lights on from 2100 to 0900 hr.).

An angled runway was used, so that the goal area at the end of the runway could not be seen from the start box. It was made of black Perspex with a clear Perspex floor. The start-box was $90 \times 150 \times 200$ mm, also made of black Perspex, with a sliding door to allow entry to the runway. The first part of the runway was $250 \times 820 \times 150$ mm wide. The second part was 30° to the first and was 380 mm long, with a clear Perspex panel located 50 mm from the inside corner of the angle. The panel was 100×100 mm.

The training programme was as follows: during pretraining, each animal was allowed to explore the runway freely for 5 min. on each of two consecutive days with a food reward available at the goal end. The reward used was a small flake of chocolate (about 500 mg). Pilot trials had shown that this reliably reinforced running behaviour in both species, without the use of any form of deprivation.

On each training day, the reward was placed at the goal end, and the animal placed in the start box. After a delay of 10 sec., the door was raised, and the animal allowed entry to the runway. Its running time was automatically recorded by photocells 100 cm from the start box and 50 cm from the goal end. Training proceeded until all animals ran to the goal without pausing to investigate any features of the runway. In practice, virtually all animals did this from the first or second day of training. Training was arbitrarily continued for 12 consecutive days (one trial per day) before testing began.

For testing, a distracting stimulus of a white card was placed inside the clear Perspex panel. The running time was recorded as before, as well as any other exploratory behaviour shown by the animal, such as direct investigation of the stimulus panel. Animals were given tests on two consecutive days (one

trial per day). Because there was possible confounding of the oestrous cycle, females of both species were only tested when in dioestrus (defined on the basis of vaginal smears). Previous work with rats had suggested that distractibility was greater at dioestrus (Birke, *et al.*, 1979), so if such a difference applies also to the two species used here, then they were being tested at the time of maximum distractibility. No attempt was made to test both at oestrus and dioestrus, because the oestrous cycle length of the species differs widely (*Mus* = 4 days, compared to 11 days for *Acomys*; Peitz, 1981). Since the mouse oestrous cycle is so short, this necessitated running the first test on the first dioestrus day. To do this, one or two additional training days were sometimes necessary to avoid testing on the day of oestrus.

Following the second testing day, animals were rerun in the runway as they had been during training (i.e., without a white panel), for two consecutive days, to determine whether there were any continuing effects on runway performance resulting from exposure to the novel stimulus. Statistical analyses then compared data from the last three days of training, the two days of testing, and the two days of posttesting, i.e., a total of 7 days of trials.

TABLE 1
RUNNING TIMES IN A RUNWAY FOR *Mus* vs *Acomys*: $M \pm SEM$

Day	Males				Females			
	<i>Mus</i>		<i>Acomys</i>		<i>Mus</i>		<i>Acomys</i>	
	<i>M</i>	<i>SEM</i>	<i>M</i>	<i>SEM</i>	<i>M</i>	<i>SEM</i>	<i>M</i>	<i>SEM</i>
Pretest								
1	5.6	0.5	3.0	0.8	6.4	0.64	3.8	0.6
2	7.3	0.6	3.9	0.9	6.5	0.7	3.2	0.4
3	5.8	0.5	3.3	0.3	5.7	0.5	4.4	1.0
Test								
4	7.4	0.5	13.1	5.8	38.6	19.4	7.4	2.9
5	8.3	1.8	19.3	9.9	27.8	8.6	11.1	3.8
Posttest								
6	9.4	1.5	2.9	0.3	9.8	0.8	3.3	0.8
7	9.6	1.4	3.0	0.2	9.5	0.7	4.2	1.1

Mean running times are shown in Table 1. Data were initially analysed using a three-way analysis of variance with repeated measures, using species and sex of animal as the variates and day of testing as the repeated measure. These analyses yielded no significant effect of sex ($F_{1,36} = 1.91$), but a significant effect of species ($F_{1,36} = 6.86$, $p < 0.05$). There was also a significant interaction of species $\times$ sex ($F_{1,36} = 7.53$, $p < 0.01$), a significant effect for the repeated measure ($F_{6,216} = 6.65$, $p < 0.01$), and a significant three-

way interaction (species $\times$ sex $\times$ repeated measure) for which $F_{6,216} = 3.15$, $p < 0.01$.

Further analyses used t tests to compare species or sex. These were only carried out for data from the first testing day, the first posttest day, and the last training day. None of the data from the sampled training day were significantly different. For the first testing day, data for *Mus* were significantly different from those for *Acomys* ($t = 2.16$, $p < 0.05$), and males were significantly different from females ($t = 2.18$, $p < 0.05$). On the first posttesting day, there was no significant difference between males and females but a significant species difference ($t = 2.6$, $p < 0.02$).

TABLE 2
NUMBER OF ANIMALS SHOWING EXPLORATORY RESPONSES
TO A DISTRACTING STIMULUS

Measure	Males		Females	
	<i>Mus</i>	<i>Acomys</i>	<i>Mus</i>	<i>Acomys</i>
Orientation	1	0	0	2
Investigation: Test 1	2	5	10	4
Turning Round	1	4	0	0
Orientation	1	1	1	1
Investigation: Test 2	4	4	9	4
Turning Round	1	1	0	0
Total responding	10	15	20	11*

* $p < 0.01$; χ^2 test. All categories were mutually exclusive.

Table 2 shows the results from recording the behaviour of the animals in the runway during testing. Again, marked species and sex differences were apparent. The summary data, for numbers of individuals responding or not were analysed using a χ^2 test, which showed a significant difference between groups ($\chi^2 = 14.8$, $p < 0.01$). Among males, somewhat more *Acomys* than *Mus* showed overt signs of responding to the stimulus directly; responses included investigating the stimulus, e.g., by sniffing its surface, orienting towards it (by direct turning of the head towards the panel), or by avoiding it and returning to the start box. Among females, however, the reverse was true. All female *Mus* responded to the change in the panel by investigation or orientation, compared with similar behaviour shown on 11 of the 20 trials for *Acomys*.

The results of this experiment suggest that there is a difference in distractibility between *Acomys* and the strain of *Mus* used here. It is not, however, in the direction predicted from our previous work with exploratory behaviour—at least for males. Male *Acomys* are more distracted by the changed stimulus than male *Mus*. This is suggested both by the increase in running

times during testing trials and by the measures of exploratory behaviour in the runway. By contrast, female *Mus* are more affected by the presence of the distracting stimulus than are female *Acomys*. Running times are greater for female *Mus* during testing; all of these animals responded directly to the stimulus by investigating it in some way.

The most dramatic species difference occurs in fact because the responses of male and female *Mus* are very different. It is not obvious why a sex difference of such magnitude should occur. If both sexes of *Mus* showed similarly frequent responses to the changed stimulus, then this might be attributed to neophobia. Cowan (1977) has shown that rat species that are commensal with humans show a marked neophobic response to novel stimuli; those species of rat which are noncommensal do not develop a similarly neophobic response. Wild *Mus musculus* would be expected to develop similar neophobia, given its proximity to human habitation. *Acomys* typically lives in rocky outcrops of arid zones of the Middle East (Dieterlen, 1962), although to what extent it might show neophobia in the wild is not known.

Neophobia, however, does not readily explain the sex difference. There is no clear functional reason why such markedly different responses to novel change should occur. In terms of mechanism it may, of course, be hormone-dependent. Distractibility has been shown to be negatively correlated with testosterone levels in young chicks (Archer, 1974) and in castrated male mice given exogenous hormone (Archer, 1977). The difference between male *Mus* and male *Acomys* may be, at least in part, due to differences in endogenous androgen levels. Although the females were all tested in dioestrus, there is a possibility that they differed in hormone levels—perhaps because of the differences between oestrous cycles—and that this might be related to the observed species difference among females. The possibility that hormonal differences underlie the differences in behaviour reported here merits further investigation.

Acomys is of interest for behavioural work because it has precocial young, unlike other *Muridae*. The young are born furred and mobile and with relatively well-developed neural organisation. Of particular interest is that the hippocampus of infant *Acomys* is highly organised (Brunjes, 1984), unlike that of altricial species. Given that at least one theory of hippocampal function has linked it with attention (Nadel & O'Keefe, 1974), a study of developmental changes in attention/distractibility in these two species could be of great interest.

Finally, it has to be said that this experiment did not support the hypothesis that male *Mus* were more distractible than male *Acomys*, which had been predicted on the basis of observation of exploratory behaviour of males of both species (Birke, *et al.*, 1985). On the other hand, an important differ-

ence between the arena used in the Birke, *et al.* (1985) study and the runway used here is that, by the time of test, the runway is relatively familiar. The arena, on the other hand, was not. The salience of a "novel" stimulus in the two situations is likely to differ, simply because the context is familiar. The lesser distractibility shown by high-androgen mice and reported by Archer (1977) might thus be specific to familiar situations, such as runways, but not be readily detectable in less familiar or predictable environments, precisely because animals do tend to explore a variety of stimuli in rapid succession. What this experiment has shown is a species and sex difference in the responses of *Mus* and *Acomys* to an unexpected, novel stimulus. It remains to be determined how familiarity with testing conditions might affect such responses.

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